

SUJATHA NATARAJAN
PROFESSIONAL EXPERIENCE

BHEL (Bharat Heavy Electronics Limited) Electronics Division

Bangalore, India

Engineering Intern

July 2024 - August 2024

- Worked on live embedded systems on Traction Loco & EMU Product and Substation Automation IED – Bay Control Unit.
- Gained hands-on experience in design and development of Tiger Sharc DSP Interface logic targeted to M/s Intel Cyclone V Soc FPGA; wrote C Programs on Micrium Ucos II RTOS based Board Support package for NXP ARM LPC Microcontroller based Hardware Controller Modules & Tiva based IO modules.

SECURDEN

Chennai, India

Product Consultant

March 2025 – July 2025

- Provided technical Product (Unified Privilege Access management (PAM), Password Vault, End-point Privilege Management) Demonstrations to clients.
- Worked on presales/sales –gave concise sales pitches, solving support and evaluation tickets.

ROCKFORT ROBOTICS

Chennai, India

Intern

August 2025 – Present

- Insight into robotic path-planning algorithms (A*, TSP) and warehouse orchestration tools using Python. Hands-on experience with differential kinematics and ROS for mobile robot navigation and motion analysis.

EDUCATION

SSN COLLEGE OF ENGINEERING

Tamil Nadu, India

Bachelor of Engineering, Electronics and Communication Engineering CGPA: 7.18/10

August 2021 - May 2025

- Member of the Photography Club, Tech Club and Music Club

PROJECTS

- **Vehicle-Vehicle Communication using an OIRS:** (IoT, Robotics, Microcontroller Programming, Optics and Reflection, Thingspeak). An elevated mirror-based V2V communication system using optical reflection for improved safety at hairpin bends and junctions.
- **IoT-based Solar Power Scarecrow:** (Arduino Programming Analog, Digital Circuit Design) An automatic solar-powered scarecrow with motorized arms facilitating crop protection from animals.
- **Anti-Sleep Alarm Eye glasses:** (Arduino IDE), Analog Circuit Design) An automatic eyeglasses setup that gives alerts when the user falls asleep, useful for night driving.
- **Online Weather Monitoring System:** (Thingspeak, Cloud, Sensor Integration, Raspberry Pi, Web Development) A Raspberry Pi-based automated weather monitoring system with sensors, map pointers, real-time data transmission to Thingspeak, and a website to present the same.
- **Physical Activity Recognition, Monitoring Rep System :**(LSTM, Mediapipe) An activity recognition and repetition counting system using video frame analysis-CNN, LSTM, and Mediapipe.

CERTIFICATIONS

National Programme on Technology Enhanced Learning (NPTEL)

- Enclosure Design of Electronic Equipment – January 2023
- Foundation of Cloud IOT – November 2022
- Natural Resource Management – December 2024

Goethe-Zertifikat A1.1

– November 2021

Zenith Hackathon (College Hackathon) Runner up Certification

– August 2022

IT TECHNICAL

- **Languages:** Python, C, and Java
- **Operating Systems:** Windows 10, 11, Arduino IDE
- **Packages:** Microsoft Word, Excel, Powerpoint, Matlab

ADDITIONAL INFORMATION

- Trained Indian Classical vocalist – Learnt Carnatic music for 10+ years, published a CD, performed in concerts.
- Amateur – Swimmer, Badminton, Table tennis, Football (represented school team)
- Languages: Fluent (Read & Write) in English, Tamil and Hindi. Basics of Spanish.